

Omprakash Reddy Vallapu

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PROFESSIONAL SUMMARY

AI/ML Engineer with 4+ years building production machine learning systems at Fortune 500 companies (UnitedHealthcare, Vanguard, JPMorgan Chase). Expert in Python, cloud infrastructure (AWS SageMaker, Azure), ML model development, LLM/RAG pipelines, and deploying scalable inference APIs. Proven track record improving model accuracy by 10–12%, cutting inference latency by 25%, and reducing operational overhead by 40%. Seeking ML Engineer, LLM Engineer, or ML Infrastructure Engineer roles.

EDUCATION

Florida Atlantic University Master of Science, <i>Computer Science</i>	Aug 2023 - May 2025 GPA: 3.9/4.0
TKR College of Engineering and Technology (affiliated with JNTUH) Bachelor of Technology, <i>Computer Science and Engineering</i>	Jan 2017 - Dec 2021 GPA: 8.0/10

SKILLS

Languages: Python, SQL, R, Java
Machine Learning: Regression, Classification, Clustering, Feature Engineering, Model Evaluation, Ensemble Methods, Scikit-learn, XGBoost
Deep Learning: Neural Networks, CNNs, RNNs/LSTMs, Transformers, TensorFlow, PyTorch
Generative AI / LLMs: Prompt Engineering, RAG (Retrieval-Augmented Generation), LLM Fine-Tuning, AI Agents, Embedding Models, LangChain, LlamaIndex, OpenAI API, Hugging Face
NLP: Text Classification, Named Entity Recognition, Sentiment Analysis, Topic Modeling
Data Engineering: ETL/ELT Pipelines, Data Cleaning, Data Wrangling, Batch & Streaming Processing, Apache Spark, Airflow
MLOps & Experimentation: MLflow, CI/CD for ML, Model Monitoring, Model Deployment, Docker, Kubernetes
Databases: SQL Server, PostgreSQL, MySQL, Snowflake, Pinecone, FAISS, ChromaDB
Cloud & DevOps: AWS (S3, EC2, Lambda, SageMaker, Glue), Azure (Azure ML, Data Factory), GCP, Git, GitHub, Azure DevOps
Full Stack / APIs: FastAPI, Flask, REST APIs
Visualization / BI: Power BI, Tableau, Matplotlib, Seaborn, Excel
Methodologies: Agile, Scrum, Requirements Gathering, Stakeholder Management, Cross-functional Collaboration

WORK EXPERIENCE

- UnitedHealthcare | AI/ML Engineer** Aug 2025 - Present
 - Built and trained machine learning models (risk scoring, readmission/cost prediction) using Python, Scikit-learn, and Pandas on large-scale healthcare claims and member data, improving prediction accuracy by 12% over baseline models.
 - Developed and deployed ML pipelines to production using Docker and cloud infrastructure, achieving 99% pipeline reliability and reducing inference latency by 25%.
 - Engineered features from structured and unstructured healthcare data (claims, EHR fields) to improve model performance, lifting F1-score by 10%.
 - Deployed machine learning models using Amazon SageMaker, Docker, and FastAPI, enabling scalable REST APIs for real-time inference and batch prediction workflows.
 - Built scalable ETL and feature engineering pipelines using AWS Glue, Lambda, S3, and Pandas to process structured and semi-structured healthcare claims data, reducing manual data preparation by 40%.
 - Automated data ingestion and preprocessing pipelines, reducing manual data preparation time by 6 hours per week.
 - Implemented model monitoring and validation processes using MLflow to track performance and flag drift over time.
 - Built REST APIs (FastAPI/Flask) to expose model predictions to internal clinical and operations applications.
 - Partnered with data engineers, clinical SMEs, and product stakeholders in an Agile/Scrum environment to define requirements and translate them into deployable ML solutions.
- Vanguard | AI/ML Engineer (CPT Employment)** Jul 2024 - Apr 2025
 - Designed and built machine learning models and Generative AI solutions supporting investment operations and client-facing analytics, partnering with product and business stakeholders to define requirements.
 - Developed and fine-tuned LLM-based applications, including Retrieval-Augmented Generation (RAG) pipelines using LangChain and vector databases (FAISS/Pinecone), reducing document search time by 30% and improving retrieval accuracy by 15%.
 - Built and maintained scalable data and ML pipelines in Python, integrating structured and unstructured data sources for model training and inference.
 - Deployed models and APIs using Docker, FastAPI/Flask, and cloud infrastructure (Azure), maintaining 99.5% uptime and reducing inference latency by 20%.
 - Implemented model monitoring and CI/CD workflows using MLflow and version control best practices to track performance drift and streamline releases.
 - Collaborated with data engineering and platform teams in an Agile/Scrum environment, participating in sprint planning, code reviews, and stakeholder demos.
- JPMorgan Chase & Co. | Data Analyst** Jun 2021 - Aug 2023
 - Analyzed financial and transactional datasets using SQL and Python/Excel to identify patterns supporting risk, fraud, and operational reporting for card services and payment operations.
 - Built and maintained automated reports and dashboards (Power BI, Tableau, Excel) used by business stakeholders for daily/weekly decision-making.
 - Performed data validation, reconciliation, and quality checks across multiple source systems to ensure accuracy of downstream reporting.
 - Wrote complex SQL queries and stored procedures to extract, transform, and aggregate data from relational databases for ad-hoc and recurring analysis.
 - Collaborated with business and technology teams in Agile ceremonies to gather requirements and prioritize analytics deliverables.
 - Identified process inefficiencies and proposed automation solutions, reducing manual effort by 5 hours per week.
 - Supported early exploratory data science initiatives, applying clustering and basic statistical techniques to customer segmentation problems.

PROJECTS

- RAG-Based Document Q&A Chatbot**
 - Built a Retrieval-Augmented Generation pipeline using LangChain, OpenAI embeddings, and a FAISS vector store, exposed via a FastAPI backend, to replace slow manual document search; reduced document lookup time by 35% and improved answer relevance by 20% through prompt engineering and chunking optimization.
- Customer Churn Prediction Model**
 - Engineered features from historical transaction/behavior data and trained classification models (Logistic Regression, Random Forest, XGBoost), tuning hyperparameters for optimal recall/precision trade-off; achieved 84% accuracy (0.85 AUC) to enable targeted retention campaigns.
- Sales/Demand Forecasting**
 - Built time-series forecasting models to improve short-term demand forecasts for inventory/staffing optimization, validated performance against historical data, and visualized results in Power BI dashboards; improved forecast accuracy by 15% versus baseline.